

Alport Syndrome and Other Familial Glomerular Syndromes

Michelle N. Rheault, Clifford E. Kashtan

ALPORT SYNDROME

Definition

Alport syndrome (AS) is an inherited disorder of basement membranes caused by mutations affecting specific proteins of the type IV (basement membrane) collagen family. The major features of AS are hematuria, progressive nephritis with proteinuria and declining kidney function, sensorineural deafness, and ocular abnormalities. Although AS is a rare disease, affecting approximately 1 in 50,000 people, it is the second most common monogenic kidney disorder, after polycystic kidney disease. AS is seen in patients of all ethnicities and nationalities and accounts for approximately 0.5% of adults and 1.7% of children with end-stage kidney disease (ESKD) in the United States.¹

Etiology and Pathogenesis

Type IV Collagen

Type IV collagen is a major constituent of basement membranes. The type IV collagen family of proteins involves six isomeric chains, designated $\alpha 1(\text{IV})$ to $\alpha 6(\text{IV})$. Each of these chains has a major collagenous domain of about 1400 residues containing the repetitive triplet sequence glycine (Gly)–X–Y, in which X and Y represent a variety of other amino acids; a C-terminal noncollagenous (NC1) domain of about 230 residues; and a noncollagenous N-terminal sequence of 15 to 20 residues.

Each type IV collagen molecule is a heterotrimer composed of three α chains. Formation of these heterotrimers is initiated by C-terminal NC1 domain interactions, accompanied by folding of the collagenous domains into triple helices. There are at least three types of type IV collagen heterotrimer: $\alpha 1(\text{IV})_2\text{--}\alpha 2(\text{IV})$, $\alpha 3(\text{IV})\text{--}\alpha 4(\text{IV})\text{--}\alpha 5(\text{IV})$, and $\alpha 5(\text{IV})_2\text{--}\alpha 6(\text{IV})$. Type IV collagen triple helices form open, nonfibrillar networks that associate with laminin assemblies through interactions mediated by nidogen to form basement membranes.

The six type IV collagen genes are arranged in pairs on three chromosomes (Fig. 48.1). The 5' ends of each gene pair are adjacent to each other, separated by sequences of varying length that contain motifs involved in transcriptional regulation.

In basement membranes there is a ubiquitous network involving the $\alpha 1(\text{IV})$ and $\alpha 2(\text{IV})$ chains; and other networks, restricted in distribution, composed of $\alpha 3(\text{IV})$, $\alpha 4(\text{IV})$, and $\alpha 5(\text{IV})$ chains, or $\alpha 5(\text{IV})$ and $\alpha 6(\text{IV})$ chains. Glomerular basement membrane (GBM) contains separate $\alpha 1\text{--}\alpha 2(\text{IV})$ and $\alpha 3\text{--}\alpha 4\text{--}\alpha 5(\text{IV})$ networks, whereas epidermal basement membranes (EBMs) contain separate networks of $\alpha 1\text{--}\alpha 2(\text{IV})$ chains and $\alpha 5\text{--}\alpha 6(\text{IV})$ chains. The $\alpha 3\alpha 4\alpha 5(\text{IV})$ network is also expressed in Bowman's capsule and distal tubule in the kidney, alveolar basement membranes, and basement membranes of the testis, eye, and ear.² A network of $\alpha 1$, $\alpha 2$, $\alpha 5$, and $\alpha 6(\text{IV})$ chains has been described in smooth muscle basement membranes.² These various networks likely have different physical and functional characteristics and

interact differently with other matrix components and adjacent cells. $\alpha 1\text{--}\alpha 2(\text{IV})$ chains are also found in mesangial matrix.

Genetics

There are three forms of AS: an X-linked form resulting from mutations at the *COL4A5* locus, primarily affecting the $\alpha 5(\text{IV})$ chain; an autosomal recessive form arising from mutations at the *COL4A3* locus or the *COL4A4* locus, affecting the $\alpha 3(\text{IV})$ and $\alpha 4(\text{IV})$ chains, respectively; and an autosomal dominant form from heterozygous mutations in *COL4A3* or *COL4A4* (Table 48.1). Approximately 80% of AS was previously thought to be inherited in an X-linked manner, with 15% autosomal recessive and 5% autosomal dominant inheritance patterns observed. With the advent of next-generation sequencing (NGS), it is clear that autosomal dominant AS (ADAS) is more common than previously recognized, accounting for approximately 20% to 30% of affected families.³

X-linked Alport syndrome. X-linked AS (XLAS) is the predominant form of the disease. Hundreds of *COL4A5* variants have been described, mostly missense mutations, splice-site mutations, and deletions of fewer than 10 base pairs. A common missense mutation involves replacement of a glycine residue in the collagenous domain of the $\alpha 5(\text{IV})$ chain by another amino acid. Such mutations are thought to interfere with the normal folding of the $\alpha 5(\text{IV})$ chain into triple helices with other $\alpha(\text{IV})$ chains.

In untreated males with *COL4A5* mutations, the risk of progression to ESKD is 50% by age 30 years, 80% by 40 years, and nearly 100% by 60 years, and most experience deafness.⁴ Genotype has a strong correlation with kidney disease progression in males with XLAS. In males with truncating variants (large deletions, nonsense mutations, truncating splicing variants, or a small mutation changing the mRNA reading frame), the risk for developing ESKD before age 30 years is 90%. Nontruncating splice-site mutations and missense mutations have a less severe renal phenotype with 70% and 50% reaching ESKD by age 30 years, respectively.^{4,5} Females with XLAS have a wide variability in disease severity, and there is no genotype-phenotype correlation.⁶ The severity of disease in a female heterozygous for a *COL4A5* mutation probably depends primarily on the relative activities of the mutant and normal X chromosomes in renal, cochlear, and ocular tissues because of random X-inactivation.⁷

Autosomal recessive Alport syndrome. Autosomal recessive AS (ARAS) arises from mutations affecting both alleles of *COL4A3* or *COL4A4*.⁸ ARAS should be suspected when an individual with typical clinicopathologic features of the disease lacks a family history, especially when a young female has severe disease, such as deafness, nephrotic syndrome, or impaired kidney function. However, sporadic cases of AS may represent de novo mutations at the *COL4A5* locus or a germline *COL4A5* mutation in the proband's mother. Most patients with ARAS develop ESKD and deafness before age 30 years, regardless of sex.⁸

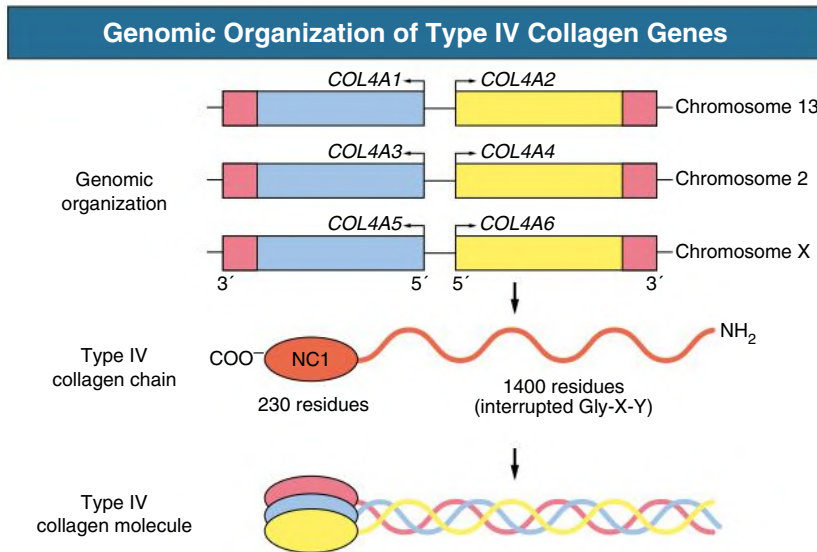


Fig. 48.1 Genomic Organization of Type IV Collagen Genes. Each type IV collagen chain is synthesized from one of the pairs of type IV collagen genes. Each type IV collagen molecule is a heterotrimer of three type IV collagen chains (See also Fig. 48.4).

TABLE 48.1 Molecular Genetics of Alport Syndrome

Inheritance	Affected Locus	Gene Product
X-linked Alport syndrome (XLAS)	<i>COL4A5</i>	$\alpha 5(\text{IV})$
X-linked + leiomyomatosis	<i>COL4A5 + COL4A6</i>	$\alpha 5(\text{IV}) + \alpha 6(\text{IV})$
Autosomal-recessive	<i>COL4A3</i>	$\alpha 3(\text{IV})$
Alport syndrome (ARAS)	<i>COL4A4</i>	$\alpha 4(\text{IV})$
Autosomal-dominant	<i>COL4A3</i>	$\alpha 3(\text{IV})$
Alport syndrome (ADAS)	<i>COL4A4</i>	$\alpha 4(\text{IV})$

Autosomal dominant Alport syndrome. Heterozygous mutations in *COL4A3* or *COL4A4* are associated with a spectrum of phenotypes, including isolated microhematuria, chronic kidney disease (CKD) with proteinuria, and ESKD. We prefer to classify all individuals with heterozygous variants in *COL4A3* or *COL4A4* as having ADAS, including those with hematuria and attenuated GBMs.⁹ Patients with progressive ADAS tend to have a slower course to ESKD than those with XLAS or ARAS.^{10,11}

COL4A3 and *COL4A4* heterozygous variants may also manifest as focal segmental glomerulosclerosis (FSGS) with or without classic AS GBM findings of thinning or thickening with lamellation. This association has been reported frequently since the initial report in 2007.^{12–14} Type IV collagen mutations are among the most common genetic mutations identified in up to 31% of adults with familial FSGS¹⁴ and up to 10% of a cohort of predominantly sporadic FSGS.¹⁵

Type IV Collagen in Alport Basement Membranes

The GBMs and tubular basement membranes of males with XLAS usually fail to express the $\alpha 3(\text{IV})$, $\alpha 4(\text{IV})$, and $\alpha 5(\text{IV})$ chains but do express the $\alpha 1(\text{IV})$ and $\alpha 2(\text{IV})$ chains (Fig. 48.2). Females with XLAS frequently exhibit mosaic expression of the $\alpha 3(\text{IV})$, $\alpha 4(\text{IV})$, and $\alpha 5(\text{IV})$ chains in GBM, whereas expression of the $\alpha 1(\text{IV})$ and $\alpha 2(\text{IV})$ chains is preserved. Most males with XLAS show no EBM expression of $\alpha 5(\text{IV})$, whereas female heterozygotes frequently display mosaicism (Fig. 48.3). Lens capsules of some males with XLAS do not express the $\alpha 3(\text{IV})$,

$\alpha 4(\text{IV})$, or $\alpha 5(\text{IV})$ chains, whereas expression of these chains appears normal in other patients.

In most patients with ARAS, GBM shows no expression of the $\alpha 3(\text{IV})$, $\alpha 4(\text{IV})$, or $\alpha 5(\text{IV})$ chains, but $\alpha 5(\text{IV})$ and $\alpha 6(\text{IV})$ are expressed in Bowman's capsule, distal thin basement membrane (TBM), and EBM (Fig. 48.4). Therefore, XLAS and ARAS may be differentiated by immunohistochemical analysis. Basement membrane expression of type IV collagen α chains appears to be normal in patients with ADAS.

These observations indicate that a mutation affecting one of the chains involved in the $\alpha 3$ – $\alpha 4$ – $\alpha 5(\text{IV})$ network can prevent GBM expression not only of that chain but also of the other two chains as well; evidence suggests that this reflects posttranslational events. Some mutant chains are unable to participate in the formation of trimers; as a result, the normal chains that are prevented from forming trimers undergo degradation. Other mutations may allow formation of abnormal trimers that are degraded before deposition in basement membranes can occur (Fig. 48.5).

Ultrastructural studies of human AS cochleae suggest that the hearing deficit may be attributable to a defect in adherence of the organ of Corti to the basilar membrane.¹⁶ The lens capsules of AS patients with anterior lenticonus exhibit thinning and dehiscence by electron microscopy (EM) indicative of mechanical failure.¹⁷

Clinical Manifestations

Kidney Involvement

Hematuria is the cardinal finding of AS. Affected males have persistent microhematuria. Many also have episodic macrohematuria during upper respiratory tract infections, usually during the first two decades of life. Hematuria has been discovered in the first year of life in affected males, in whom it is probably present from birth. Males who are free of hematuria during the first 10 years of life are unlikely to be affected.

More than 90% of females with XLAS have persistent or intermittent microhematuria, but about 5% of obligate heterozygotes never manifest hematuria.⁶ Hematuria is persistent in both males and females with ARAS. About 50% of individuals with heterozygous variants in *COL4A3* or *COL4A4* mutations have hematuria.

Proteinuria is usually absent early in life but develops eventually in all males with XLAS and in males and females with ARAS. Proteinuria increases progressively with age and may result in nephrotic syndrome.

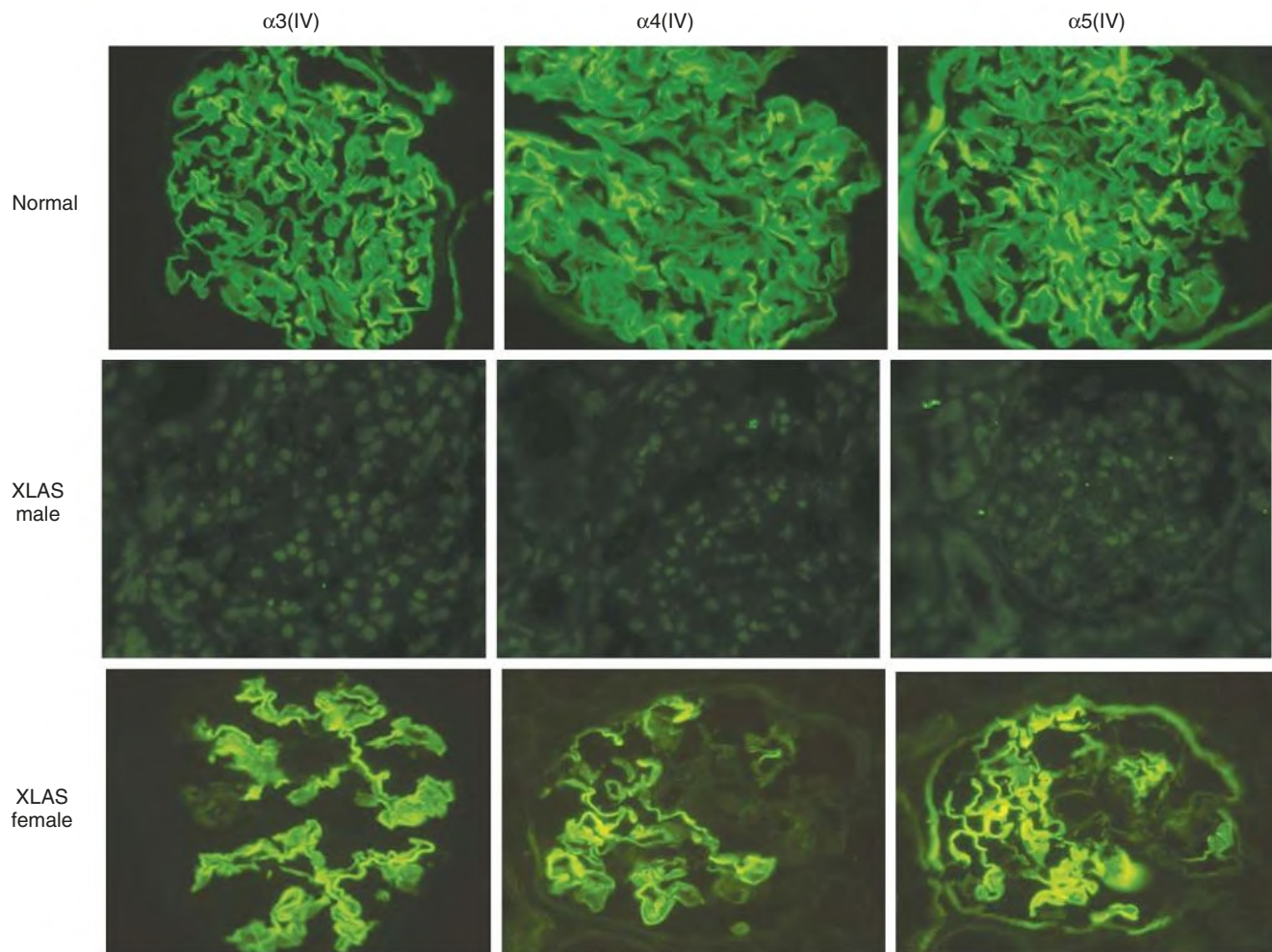


Fig. 48.2 Immunohistochemistry of Glomerular Basement Membrane (GBM) in X-Linked Alport Syndrome. In a normal individual (*top row*), GBM stains strongly for the $\alpha 3(\text{IV})$, $\alpha 4(\text{IV})$, and $\alpha 5(\text{IV})$ chains of type IV collagen. Staining of GBM of an affected male is negative for each of these chains (*middle row*), whereas an affected female shows mosaic immunoreactivity (*bottom row*).

Proteinuria develops eventually in at least 75% of heterozygous females⁶ and in an uncertain percentage of individuals with heterozygous variants in *COL4A3* or *COL4A4*. Hypertension also increases in incidence and severity with age and is more likely in affected males than affected females with XLAS, but there are no sex differences in hypertension frequency in ARAS.

ESKD develops in all affected males with XLAS at a rate determined primarily by the nature of the underlying *COL4A5* mutation.⁴ Thus, the rate of progression is fairly constant among affected males within a particular family, but there is significant interkindred variability. Significant intrakindred variability in the rate of progression to ESKD has been reported in some families with missense *COL4A5* mutations.

Progression to ESKD in females with XLAS was considered an unusual event until a 2003 European study of several hundred females with XLAS found that 12% developed ESKD before age 40 years (vs. 90% of males with XLAS), increasing to 30% by 60 years and 40% by 80 years.⁶ A confirmatory European registry study showed a 15% prevalence of ESKD in heterozygous females.¹⁸ The risk for ESKD is significantly increased in heterozygotes with proteinuria. The outcome of XLAS in females is presumed to depend on the relative activities of the normal and mutant X chromosomes, but other unknown factors also likely influence outcome.⁷ Macrohematuria in childhood, nephrotic

syndrome, and the finding of diffuse GBM thickening by EM are risk factors for CKD in affected females.¹⁹ Sensorineural deafness and anterior lenticonus also suggest an unfavorable outcome in affected females. Both males and females with ARAS appear likely to progress to ESKD during the second or third decade of life.⁸

Cochlear Defects

Deafness is frequently but not universally associated with the Alport kidney lesion, occurring in about 90% of males and 25% to 30% of females with XLAS.^{4,6} In some families with Alport nephropathy and apparently normal hearing, deafness may occur late and may be very slowly progressive.

Hearing loss in AS is never congenital and usually becomes apparent by late childhood to early adolescence in males with XLAS and in both males and females with ARAS. Hearing loss is present in 50% of males with XLAS by approximately age 15 years, 75% by 25 years, and 90% by 40 years.⁴ Hearing loss is less common in females with XLAS, with 10% of XLAS affected by age 40 years and about 20% by 60 years.⁶ Hearing loss is also common in ARAS with approximately 40% to 66% of individuals affected⁸ but less common in ADAS with only 2% to 13% affected.^{10,11} Hearing impairment in members of families with AS is always accompanied by evidence of kidney involvement. In its early stages, the hearing deficit is detectable only by audiometry, typically

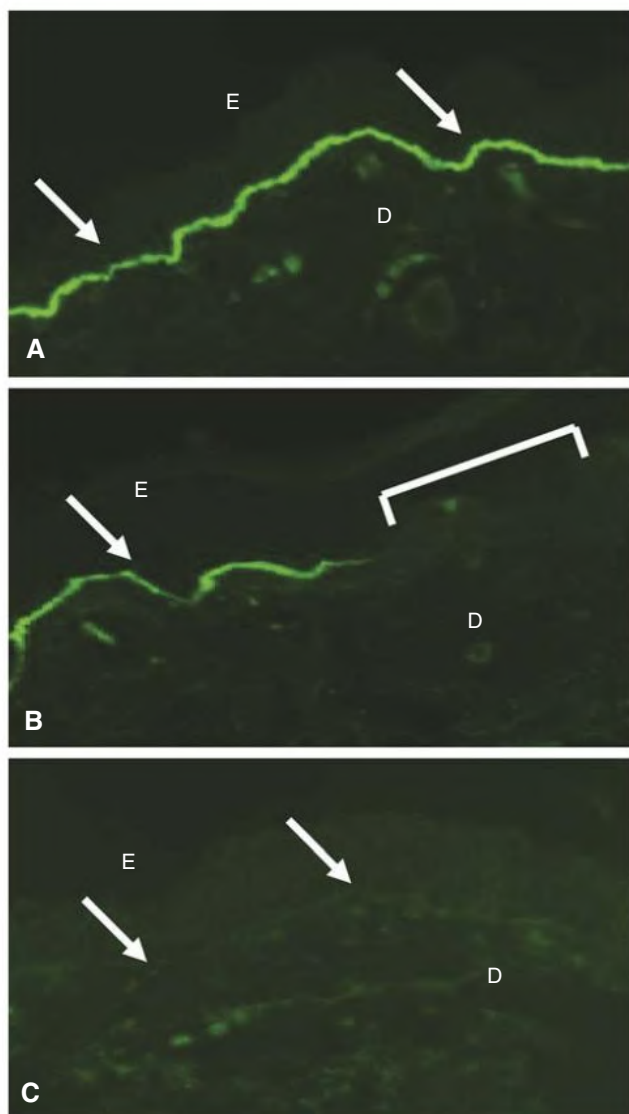


Fig. 48.3 Immunohistochemistry of Epidermal Basement Membrane (EBM) in X-Linked Alport Syndrome. (A) In a normal male, EBM shows strong staining for $\alpha 5(\text{IV})$ at the dermoepidermal junction (arrows) between dermis (D) and epidermis (E). (B) In an affected female, EBM shows mosaic staining (arrow); the white bracket identifies a length of EBM negative for $\alpha 5(\text{IV})$. (C) In affected males, staining of EBM (arrows) for $\alpha 5(\text{IV})$ is absent.

with bilateral reduction in sensitivity to tones in the range of 2000 to 8000 Hz. In affected males, the deficit extends progressively to other frequencies, including those of conversational speech.

Ocular Defects

Ocular defects occur in 30% to 40% of males with XLAS and in about 15% of XLAS females.^{4,6} *Anterior lenticonus*, a cone-shaped distortion of the anterior surface of the lens, is virtually pathognomonic of AS, occurs in about 15% of males with XLAS, and is almost entirely restricted to AS families with progression to ESKD before age 30 years and deafness.⁴ Anterior lenticonus is absent at birth, usually appearing during the second to third decade of life after the onset of CKD and is bilateral in 75% of patients (Fig. 48.6A). The spectrum and frequencies of ocular lesions appear to be similar in XLAS and ARAS.²⁰ Ocular lesions are almost never observed in ADAS.^{10,11}

Another common ocular manifestation of AS is a maculopathy, with whitish or yellowish flecks or granulations in a perimacular distribution. This is present in 50% to 60% of males with XLAS, in males and females with ARAS, and in approximately 15% of females with XLAS²⁰ (see Fig. 48.6B). In most individuals the maculopathy does not appear to be associated with any visual abnormalities, although giant macular holes causing visual impairment have been reported in some subjects.^{20a}

Corneal endothelial vesicles (posterior polymorphous dystrophy) have been observed in AS and may indicate defects in the Descemet membrane, the basement membrane underlying the corneal endothelium. Recurrent corneal erosion in AS has been attributed to alterations of the corneal EBM.

Leiomyomatosis

The association of AS with leiomyomatosis of the esophagus and tracheobronchial tree has been reported in about 30 families. Affected females typically also have genital leiomyomas, with clitoral hypertrophy and variable involvement of the labia majora and uterus. Bilateral posterior subcapsular cataracts also occur frequently in affected individuals. Symptoms usually appear in late childhood and include dysphagia, postprandial vomiting, retrosternal or epigastric pain, recurrent bronchitis, dyspnea, cough, and stridor. This syndrome typically arises from a contiguous gene deletion on the X chromosome involving exon 1 of *COL4A5*, the common promoter region that regulates gene expression of *COL4A5* and *COL4A6*, and the first 2 exons of the adjacent *COL4A6* gene.²¹ The genotype-phenotype relationship in this disorder is uncertain because deletions in this region may occur without associated leiomyomas and, conversely, some families with XLAS and leiomyomas do not have deletions involving the common promoter region and *COL4A6*.²²

Hematologic Defects

An autosomal dominant syndrome of hereditary nephritis, deafness, and megathrombocytopenia called *Epstein syndrome* has been described in a small number of families. Families with Fechtner syndrome exhibit these features and leukocyte inclusions (May-Hegglin anomaly). Both Epstein and Fechtner syndrome arise from mutations in nonmuscle myosin heavy-chain IIA (*MYH9*).²³ Although in some patients the ultrastructural changes in GBM resemble those of AS, basement membranes of these patients do not have abnormal expression of type IV collagen α chains. Therefore, Epstein and Fechtner syndromes are best considered distinct forms of hereditary nephritis rather than variants of AS.

Arterial Abnormalities

Aneurysmal dilation of the thoracic and abdominal aorta and smaller arterial vessels has been described in a small number of males with AS.²⁴

Kidney Pathology

There are no pathognomonic lesions on light microscopy (LM) or immunofluorescence (IF) in AS. In affected males, biopsies obtained before 5 years of age typically show no LM changes. Mesangial hypercellularity and matrix expansion are typically observed in older children and adolescents. Glomeruli of affected males eventually show FSGS, and interstitial fibrosis and tubular atrophy are often found in affected males older than 10 years.²⁵ LM findings in affected females correlate with proteinuria and kidney function; an affected female of any age who has isolated microhematuria is likely to have little or no abnormality by LM. Indirect IF of type IV collagen α -chain expression in renal or skin basement membranes can be diagnostic (see earlier discussion).

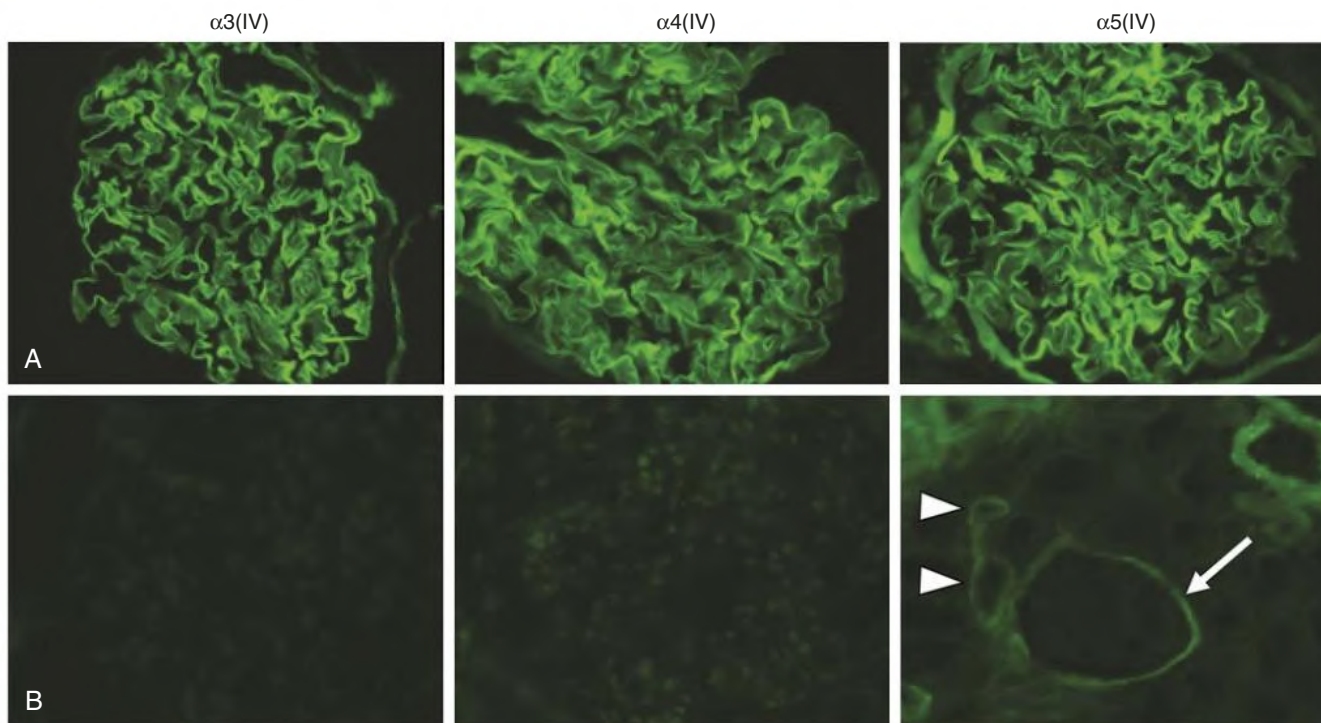


Fig. 48.4 Immunohistochemistry of Kidney in a Patient With Autosomal Recessive Alport Syndrome. (A) Normal glomerular basement membrane (GBM) and Bowman's capsule staining for $\alpha 3(\text{IV})$, $\alpha 4(\text{IV})$, and $\alpha 5(\text{IV})$. (B) Patient shows no GBM staining, but $\alpha 5(\text{IV})$ is present in Bowman's capsule (arrow) and distal tubular basement membranes (arrowheads).

EM frequently reveals diagnostic abnormalities. In most patients with AS there is variable thickening, thinning, basket weaving, and lamellation of the GBM (Fig. 48.7). The thick segments measure up to 1200 nm in depth, usually have irregular outer and inner contours, and are found more frequently in males than in females. The lamina densa is transformed into a heterogeneous network of membranous strands, which enclose clear electron-lucent areas that may contain round granules of variable density measuring 20 to 90 nm in diameter. There are variable degrees of epithelial foot process fusion.

Not all Alport kindreds demonstrate these characteristic ultrastructural features. Thick, thin, normal, and nonspecifically altered GBMs have all been described. Affected young males, heterozygous females at any age, and, on occasion, affected adult males may have diffusely attenuated GBM measuring as little as 100 nm or less in thickness rather than the pathognomonic lesion. There are a number of families reported with features of hereditary FSGS with or without the classic AS basement membrane lesion associated with mutations in *COL4A3* or *COL4A4*.¹³ These findings highlight the utility of detailed genetic evaluation to avoid misdiagnosis.

Diagnosis and Differential Diagnosis

Fig. 48.8 summarizes the evaluation of patients with hematuria and a positive family history. AS should be included in the initial differential diagnosis of patients with persistent microhematuria after excluding structural abnormalities of the kidneys or urinary tract (see Chapter 63). The presence on EM of diffuse thickening and multilamellation of the GBM predicts a progressive nephropathy, regardless of family history. However, in a patient with a negative family history, EM cannot differentiate de novo XLAS from ARAS. In some patients, the biopsy findings may be ambiguous, particularly in females and young patients of either sex. Furthermore, families with progressive nephritis and *COL4A5* mutations in association with GBM thinning have been

described, indicating that the classic Alport GBM lesion is not present in all Alport kindreds.

If an undiagnosed patient has hematuria and multiple relatives with hematuria, who should undergo kidney biopsy? The natural history of the AS kidney lesion suggests that older, male individuals are more likely to exhibit diagnostic ultrastructural GBM abnormalities. In families with a firm diagnosis of AS established, evaluation of individuals with newly recognized hematuria can be limited to ultrasound of the kidneys and urinary tract to exclude coincidental tumor or structural anomalies of the urinary tract.

Absence of the $\alpha 3$, $\alpha 4$, and $\alpha 5$ chains of type IV collagen from GBM and distal TBM is unique to AS (Table 48.2). Expression of $\alpha 5(\text{IV})$ in the EBM detected by IF may be informative, but apparently normal expression of type IV collagen α chains in basement membranes does not exclude the diagnosis of AS. Although mosaic expression of $\alpha 5(\text{IV})$ is diagnostic of a heterozygous *COL4A5* variant, a normal result does not exclude heterozygosity.

A firm histologic diagnosis of AS cannot always be established, or it may not be possible to determine the mode of transmission, despite careful evaluation of the pedigree and application of the full range of histologic methods. In these situations, genetic analysis by NGS or whole exome sequencing (WES) provides information essential for determining prognosis and guiding genetic counseling. Genetic analysis for AS is widely available in commercial and research laboratories and ideally should be pursued in all patients with suspected or biopsy-confirmed AS to establish the genotype and facilitate cascade testing of at-risk family members.

Genetic analysis using conventional Sanger sequencing identifies 80% to 90% of males with *COL4A5* mutations. NGS and WES allow for simultaneous evaluation of *COL4A3*, *COL4A4*, and *COL4A5*.²⁶ Identification of a specific genotype can provide valuable prognostic information about the risks for kidney disease progression and

Assembly of Type IV Collagen Heterotrimers in Health and in Alport Syndrome

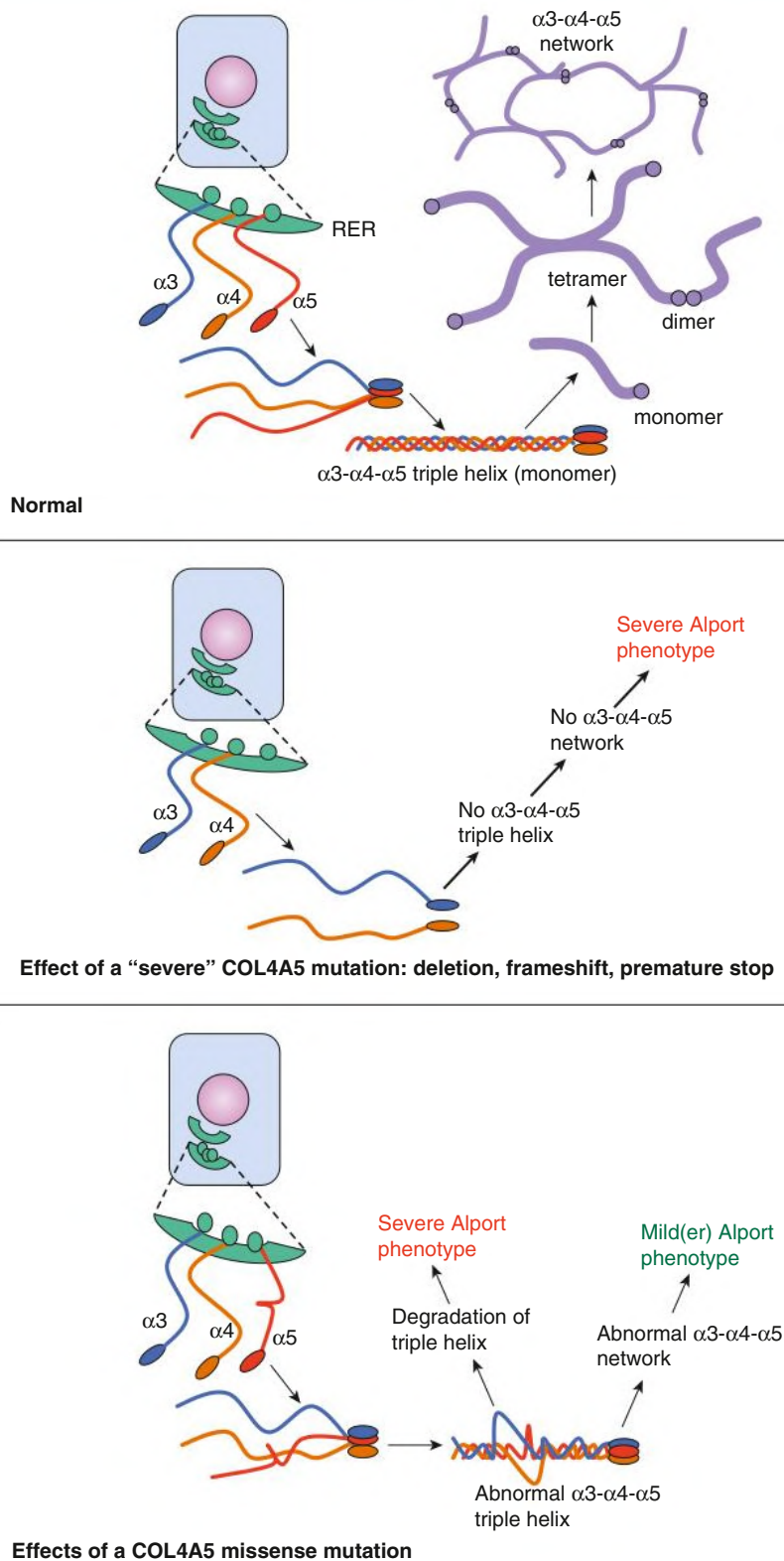


Fig. 48.5 Assembly of type IV collagen heterotrimers in health and in Alport syndrome. *RER*, Rough endoplasmic reticulum.

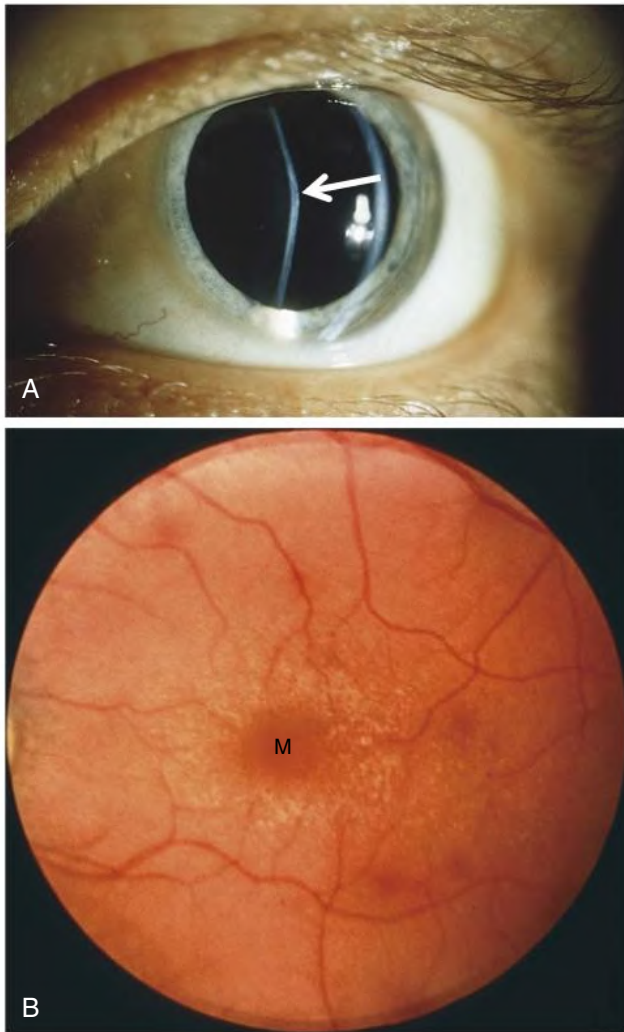


Fig. 48.6 Ocular Abnormalities in Alport Syndrome. (A) Anterior lenticonus shown by slit-lamp ophthalmoscopy. The anterior surface of the lens is cone shaped (arrow marks apex of the cone), rather than the normal smooth ellipse. (B) Perimacular flecks. Note the white flecks surrounding the macula (M). (From Flinter FA. Disorders of the basement membrane: Hereditary nephritis. In: Morgan SH, Grunfeld JP, eds. *Inherited Disorders of the Kidney*. Oxford University Press; 1998.)

associated hearing loss and eye findings in an individual patient.⁴ Once a new diagnosis of AS is made, all potentially affected family members including females should be screened for hematuria to identify those who may be at risk for progressive kidney disease, and targeted mutation analysis can be offered to at-risk individuals.²⁷

Natural History

Microhematuria, the first and invariable kidney manifestation of AS, probably reflects GBM thinning with focal ruptures because of defective expression of the $\alpha3$ – $\alpha4$ – $\alpha5$ (IV) network. Proteinuria likely results from altered podocyte interactions with the abnormal GBM, and declining kidney function is attributable to profibrotic processes in glomeruli and in the tubulointerstitial compartment. What factors initiate and drive the progression of Alport nephropathy to ESKD, and how are these factors influenced by *COL4* genotype?

Reduction in the quantity of the $\alpha3$ – $\alpha4$ – $\alpha5$ (IV) network in GBM, as likely occurs in subjects with heterozygous variants in *COL4A3* or *COL4A4*, probably has consequences different from complete loss of this network, as occurs in most males with XLAS and most patients

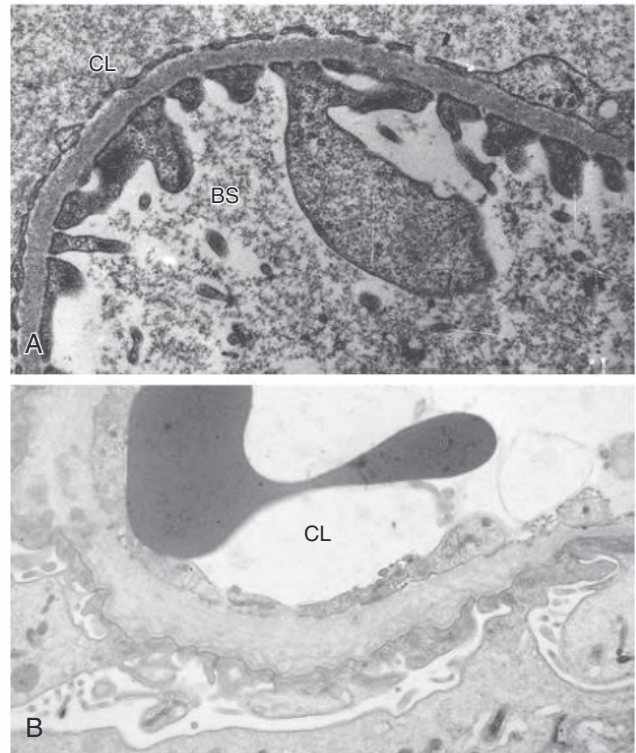


Fig. 48.7 Kidney Biopsy in Alport Syndrome. (A) Normal glomerular capillary wall. (B) Glomerular capillary wall from a patient with Alport syndrome, at the same magnification. Note the thickening of the glomerular basement membrane (GBM), the splitting of the lamina densa into multiple strands, and the marked irregularity of the epithelial aspect of the GBM in the patient with Alport syndrome. BS, Bowman's space; CL, Capillary lumen.

with ARAS. The normal transition from the $\alpha1$ (IV)₂– $\alpha2$ (IV)₁ network of nascent glomeruli to the $\alpha3$ (IV)– $\alpha4$ (IV)– $\alpha5$ (IV) network of mature glomeruli fails to occur in AS, and $\alpha1$ (IV) and $\alpha2$ (IV) chains accumulate in Alport glomeruli as the disease progresses.^{28,29} Alport GBM also shows overexpression of other matrix proteins normally absent from GBM or present in scant quantities, including type V collagen, type VI collagen, laminin $\alpha2$ chain, and fibronectin. These alterations in GBM composition are unique to AS.^{28,30} Both glomerular endothelial cells and podocytes appear to contribute to the accumulation of these proteins in Alport GBM. Recently mesangial filopodial invasion of the GBM with deposition of ectopic laminin chains was proposed as another mechanism of matrix accumulation.³¹ Alterations in glomerular extracellular matrix are accompanied by changes in glomerular cell behavior, including expression of transforming growth factor $\beta1$, integrins, and matrix metalloproteinases. Activation of fibrogenic pathways in the renal interstitium presumably represents a downstream consequence of glomerular disease.

Treatment

Current recommendations for delaying ESKD in patients with AS focus on the early introduction of renin-angiotensin-aldosterone system (RAAS) blockade to suppress urinary protein excretion.^{27,32,33} The use of RAAS blockade is based on experimental and clinical evidence as well as practical considerations. Angiotensin-converting enzyme (ACE) inhibition lengthens survival in mice with ARAS, doubling survival when begun before the onset of proteinuria. ACE inhibition begun after the onset of proteinuria also improves survival in Alport mice, to a lesser degree. The effect of

Evaluation of Patient with Hematuria and a Positive Family History

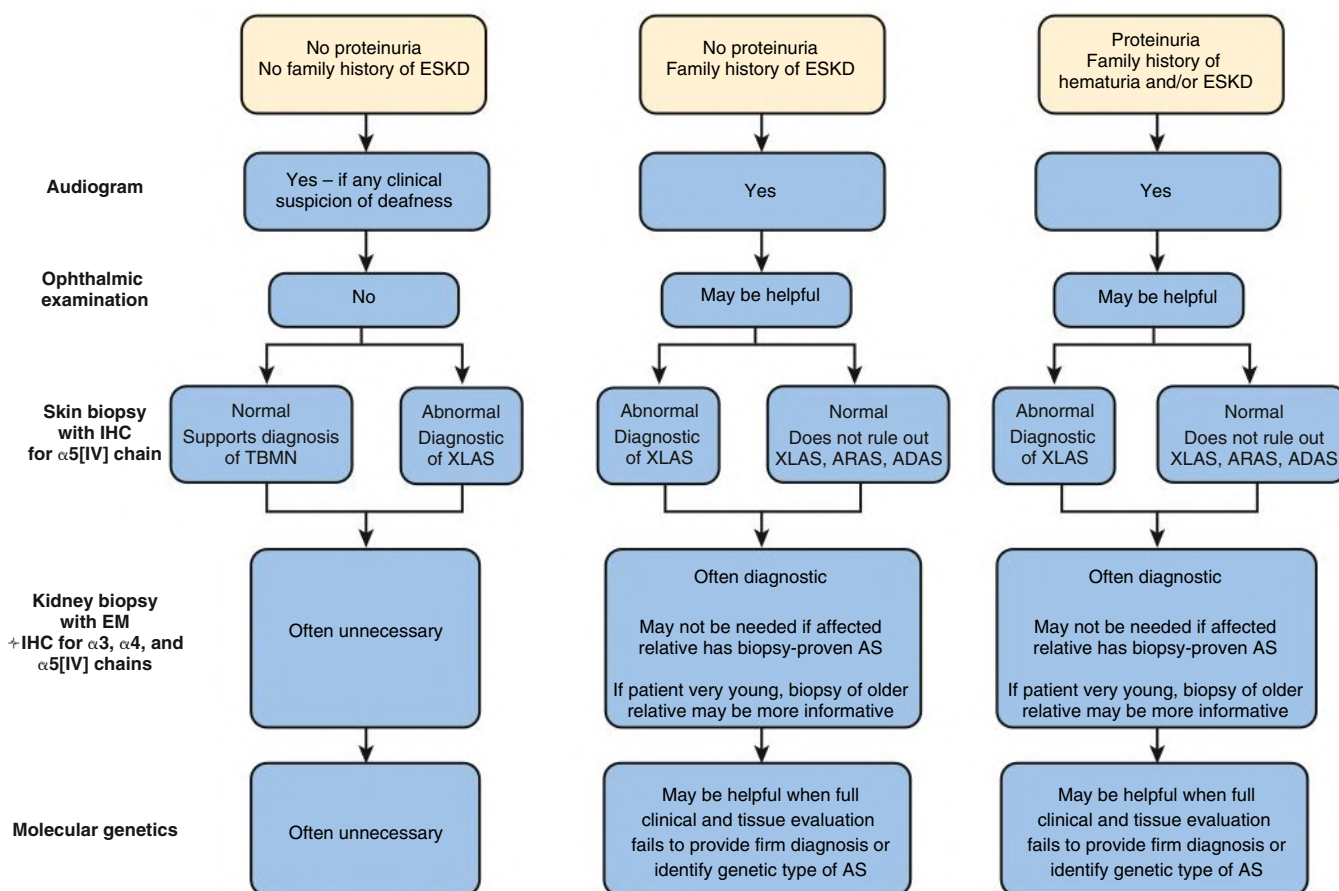


Fig. 48.8 Evaluation of patient with hematuria and positive family history. ADAS, Autosomal-dominant Alport syndrome; ARAS, autosomal recessive Alport syndrome; AS, Alport syndrome; EM, electron microscopy; ESKD, end-stage kidney disease; IHC, immunohistochemistry; TBMN, thin basement membrane nephropathy; XLAS, X-linked Alport syndrome.

ramipril on survival was superior to that of candesartan in a murine comparison study.

Uncontrolled clinical studies in AS have shown that RAAS blockade can reduce proteinuria. In a large multicenter, randomized, double-blind study comparing losartan with placebo or amlodipine in children with AS and proteinuria, there was a significant reduction in proteinuria with losartan over 12 weeks.³⁴ A 3-year extension of this study showed comparable efficacy of enalapril and losartan in reducing proteinuria.³⁵ A retrospective review of treated Chinese children with AS similarly showed a decline in proteinuria with ACE inhibition that was sustained over 5 years of follow-up.³⁶

Retrospective data from the European Alport Registry indicates that ACE inhibitor therapy initiated while glomerular filtration rate (GFR) is still normal delays ESKD by years.³⁷ This also has been shown in pairs of affected brothers discordant for ACE inhibitor therapy with a delay in ESKD of a median of 13 years in the earlier treated brother.³⁷ The beneficial effect of ACE inhibition is observed in males with XLAS because of truncating and nontruncating variants in *COL4A5*.⁵ ACE inhibitors can be used safely in children with CKD at doses that achieve suppression of urinary protein excretion.³⁸ RAAS blockers are relatively inexpensive and widely available, so in theory any individual in the world with AS should be able to receive treatment.

Consensus recommendations for the management of AS children include (1) early screening for hematuria in at-risk children, (2) regular determination of urinary protein excretion on diagnosis, and (3) initiation of RAAS blockade once overt proteinuria develops.³³ A recent update of these guidelines recommends consideration of treatment at the time of diagnosis in males with XLAS and males and females with ARAS, even if urine albumin levels are normal. A randomized controlled trial (RCT) examining the effect of RAAS blockade on the transitions from isolated hematuria to microalbuminuria to overt proteinuria showed that RAAS blockade delayed these transitions.³⁹ Prospective trials of add-on therapies such as bardoxolone methyl and anti-microRNA-21 are in progress.

Transplantation

Kidney transplantation is typically successful in patients with AS who reach ESKD with graft survival equivalent to that in patients with other diagnoses.⁴⁰ However, anti-GBM glomerulonephritis involving the kidney allograft is a rare but dramatic manifestation of AS, occurring in 2% to 3% of male patients with AS who undergo transplantation and typically presents in the first year (see Chapter 25).

Are females who are heterozygous for *COL4A5* mutations suitable kidney donors? Clearly, those with proteinuria, hypertension, or reduced GFR should not donate, nor should females with hearing

TABLE 48.2 Immunostaining for Type IV Collagen in Alport Syndrome

Type IV Collagen Group	Glomerular Basement Membrane	Bowman's Capsule	Distal Tubular Basement Membrane	Epidermal Basement Membrane
Normal (Males and Females)				
$\alpha 3(\text{IV})$	Present	Present	Present	Absent
$\alpha 4(\text{IV})$	Present	Present	Present	Absent
$\alpha 5(\text{IV})$	Present	Present	Present	Present
X-Linked (Males)^a				
$\alpha 3(\text{IV})$	Absent	Absent	Absent	Absent
$\alpha 4(\text{IV})$	Absent	Absent	Absent	Absent
$\alpha 5(\text{IV})$	Absent	Absent	Absent	Absent
X-Linked (Females)^b				
$\alpha 3(\text{IV})$	Mosaic			Absent
$\alpha 4(\text{IV})$	Mosaic			Absent
$\alpha 5(\text{IV})$	Mosaic			Mosaic
Autosomal Recessive (Males and Females)^a				
$\alpha 3(\text{IV})$	Absent	Absent	Absent	Absent
$\alpha 4(\text{IV})$	Absent	Absent	Absent	Absent
$\alpha 5(\text{IV})$	Absent	Present	Present	Present

^aIn some Alport kindreds, staining of basement membranes for type IV collagen chains is entirely normal. Therefore, a normal result does not exclude a diagnosis of X-linked Alport syndrome.

^bSome heterozygous females have normal basement membrane immunoreactivity for type IV collagen chains. Therefore, a normal result does not exclude the carrier state.

loss. A case series of heterozygous donors showed new-onset hypertension, proteinuria, and decline in GFR after kidney donation in most females.⁴¹ Given the recent finding that 30% to 40% of heterozygous females may eventually develop ESKD, the risk that a heterozygous donor will ultimately develop significant reductions in GFR must be higher than for the usual kidney donor. However, a common clinical scenario is a mother choosing to donate to her son with AS, and the choices of the whole family should be considered.

HEREDITARY ANGIOPATHY WITH NEPHROPATHY, ANEURYSMS, AND CRAMPS (HANAC SYNDROME)

An autosomal dominant hereditary angiopathy associated with nephropathy, aneurysms, and muscle cramps (HANAC syndrome) is caused by missense mutations in the *COL4A1* gene that allow for expression of an abnormal $\alpha 1(\text{IV})$ chain.⁴² Findings in affected individuals include microhematuria and macrohematuria, mild reduction in GFR, and kidney cysts. Retinal arteriolar tortuosity and retinal hemorrhage are common in affected individuals, as are intracranial aneurysms and leukoencephalopathy. Some affected individuals have elevated serum creatine kinase and muscle cramps.

Kidney biopsy in affected individuals with hematuria shows no abnormalities of GBM structure or type IV collagen expression. However, basement membranes of Bowman's capsules, tubules, and interstitial capillaries may exhibit irregular thickening, splitting into multiple layers and focal interruptions.

FABRY DISEASE (ANDERSON-FABRY DISEASE)

Definition

Fabry disease is caused by hereditary deficiency of the enzyme α -galactosidase A (α -Gal A), resulting in the intracellular accumulation of neutral glycosphingolipids with terminal α -linked galactosyl moieties (Fig. 48.9). Clinically, this leads to progressive CKD, pain crises,

sweating abnormalities, vascular cutaneous lesions, and cardiac and eye abnormalities.

Etiology and Pathogenesis

More than 1000 mutations causing Fabry disease have been identified in *GLA*, the gene for α -Gal A, which is located on the X chromosome. Most of the described mutations are associated with the classic Fabry phenotype, in which there is less than 3% α -Gal A activity with multi-system involvement. Certain missense mutations with greater than 3% α -Gal A activity have been identified in patients with a mild phenotype limited to cardiac abnormalities. Prevalence of the classic form is up to 1 in 22,570 males by newborn screening studies and 1 in 1390 males for the males with milder phenotype.⁴³

Glycosphingolipids are normal constituents of the plasma membrane, the membranes of intracellular organelles, and circulate in association with apolipoproteins. The neutral glycosphingolipids that accumulate in Fabry disease are identical to those found in normal tissue. All tissues except red blood cells accumulate globotriaosylceramide (Gb3), with the highest concentrations found in the diseased kidney.

Clinical Manifestations and Pathology

Fabry disease is a multisystem disorder, with prominent and potentially devastating involvement of the kidneys, heart, and peripheral and central nervous systems. As expected for an X-linked disorder, severe clinical manifestations occur in hemizygous males, whereas heterozygous females have a variable but typically less severe course. In affected males, the initial features of the disease are seen in childhood and early adolescence and consist of paresthesias and pain in the hands and feet with episodic pain crises. The course of the disease is variable but usually leads to ESKD in the third to sixth decade. There is risk of premature death because of myocardial or cerebral infarctions. Severe Fabry disease in a female reflects extensive inactivation of the X chromosome carrying the normal α -Gal A allele.⁴⁴

Kidney Involvement

Although the earliest manifestation of kidney involvement is a urinary concentrating defect, the nephropathy of Fabry disease typically manifests as mild to moderate proteinuria, sometimes with microhematuria, beginning in the third decade, although proteinuria in childhood is reported. Nephrotic syndrome is unusual. Urinary oval fat bodies, with a Maltese cross configuration when viewed with a polarizing microscope, are a result of the large amounts of glycosphingolipid in the urine (see Chapter 4, Fig. 4.2B). Deterioration of GFR is gradual, with hypertension and ESKD developing by the fourth or fifth decade of life. Heterozygous females typically have mild kidney involvement but up to 10% may develop ESKD.

Glomerular visceral epithelial cells are enlarged and packed with small, clear vacuoles that represent glycosphingolipid material (Fig. 48.10). Vacuoles also may be seen in parietal epithelial cells and the epithelial cells of the distal convoluted tubule and loop of Henle, but only rarely in mesangial cells, glomerular endothelial cells, or proximal tubular epithelial cells. There is progressive segmental and global glomerulosclerosis. Vacuoles are also observed in endothelial cells and smooth muscle cells of arterioles and arteries.

On EM, there are abundant inclusions within lysosomes, particularly within visceral epithelial cells (Fig. 48.11). The inclusions (myelin figures) are typically round, involving concentric layers of dense material separated by clear spaces. The layers may be arranged in parallel (zebra bodies). Detachment of visceral epithelial cells from the underlying basement membrane may be observed. Inclusions are also observed in heterozygous females, although usually in smaller numbers than in affected males. Typical inclusions may be noted in excreted renal tubular cells.

The progression of Fabry nephropathy to ESKD probably reflects two parallel processes. Visceral epithelial cell dysfunction, which results in proteinuria, is followed by visceral epithelial cell detachment and necrosis, leading to capillary loop collapse and segmental sclerosis. Simultaneously, progressive impairment of arteriolar flow may develop as enlarging endothelial cells impinge on vascular lumina, resulting in ischemic glomerular damage.

Heart Defects

Glycosphingolipid accumulation in coronary artery endothelial cells and in the myocardium results in coronary artery narrowing, which may lead to angina, myocardial infarction, or congestive heart failure. Left ventricular hypertrophy (LVH) may be an early finding. Arrhythmias resulting from infiltration of the conduction system and valvular lesions also may occur. Certain missense mutations affecting α -Gal A may present as isolated LVH.

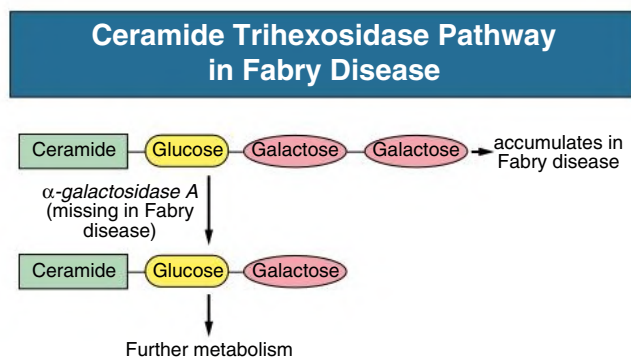


Fig. 48.9 Ceramide Trihexosidase Pathway in Fabry Disease. α -Galactosidase A deficiency leads to tissue accumulation of trihexosylceramide.

Nervous System

Autonomic dysfunction is a prominent feature of Fabry disease, typically manifested by hypohidrosis, acral paresthesias and pain, and altered intestinal motility. Cerebrovascular symptoms (secondary to narrowing of the vascular lumen from accumulation of glycosphingolipid in vascular endothelial cells) tend to appear during the fourth decade and include hemiparesis, vertigo, diplopia, dysarthria, nystagmus, nausea and vomiting, headache, ataxia, and memory loss. The vertebrobasilar circulation is preferentially involved. Symptoms are often recurrent. Life-threatening intracerebral hemorrhage and infarction are not unusual. Dementia arising from glycosphingolipid accumulation in small cerebral blood vessels also has been described.

Skin

Angiokeratomas usually appear during the second decade of life and can be the earliest presentation of Fabry disease, presenting as dark-red macules or papules of variable size (Fig. 48.12), originally known as *angiokeratoma corporis diffusum*. Typical locations include the lower trunk, buttocks, hips, genitalia, and upper thighs. The number of lesions varies from none up to 40 and increase in size and number.⁴⁵ On histologic examination, angiokeratomas consist of dilated small veins in the upper dermis, covered by hyperkeratotic epidermis. Telangiectasias may be noted, especially behind the ears.

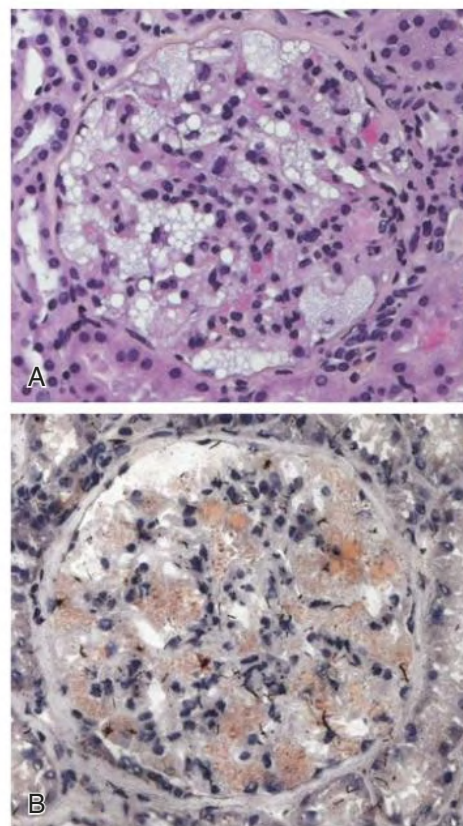


Fig. 48.10 Light Microscopy of a Kidney Biopsy Specimen in Fabry Disease. Glomerular epithelial cell glycosphingolipid deposition demonstrated by vacuolation on hematoxylin-eosin staining (A; magnification $\times 20$; B, oil red O staining $\times 20$). (Courtesy Dr. Paolo Menè and Dr. Antonella Stoppacciaro, University of Rome.)

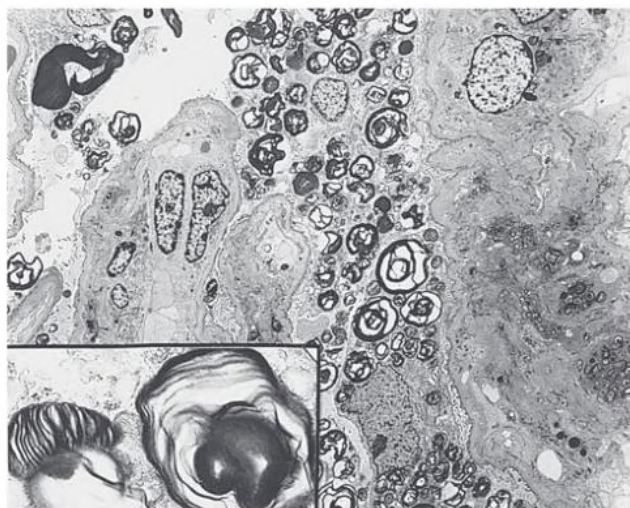


Fig. 48.11 Electron Micrograph of a Kidney Biopsy Specimen in Fabry Disease. Glycosphingolipid is deposited in cytoplasmic vacuoles in glomerular visceral epithelial cells. *Inset*, Cytoplasmic vacuoles contain electron-dense material in parallel arrays (zebra bodies) and in concentric whorls (myelin figures). (Courtesy Dr. J. Carlos Manivel.)

Eyes

Characteristic corneal opacities are common in both males and females with Fabry disease, present in approximately 75% of affected males and females.⁴⁶ These lesions, termed *verticillata*, are identified by slit-lamp examination and are whorls of whitish discoloration that radiate from the center of the cornea. Cataracts (23% of males and 10% of females) and dilated conjunctival or retinal vessels may be observed.⁴⁶

Lungs

Dyspnea and cough are common in males with Fabry disease, often with obstructive features on spirometry. This may be a consequence of fixed narrowing of the airways caused by glycosphingolipid accumulation.

Diagnosis

Diagnosis of affected males usually can be made clinically, with the additional information from slit-lamp eye examination to identify corneal opacities. The diagnosis should be confirmed by demonstrating decreased (<25%) or absent α -Gal A activity in serum, leukocytes, cultured skin fibroblasts, or tissue. Atypical variants may have enzyme activity up to 35% of normal. Heterozygous females may have very low (<25%) α -Gal A activity similar to that in males or may have more normal levels, making measurement of enzyme activity an insensitive way of diagnosing carriers. If enzyme levels are equivocal, alternative approaches to diagnosis include careful slit-lamp eye examination, measurement of urinary ceramide digalactoside and trihexoside, kidney biopsy, and genetic testing. Genetic testing for Fabry disease is widely available. Identification of carriers is particularly relevant when family members are being considered as living kidney donors.

Fabry disease should be considered in patients with unexplained ESKD, especially if LVH is present or there is a history of stroke or non-diabetic paresthesia. Fabry disease accounts for 0.1% to 0.2% of patients on hemodialysis and 0.9% of patients with hypertensive LVH.⁴⁷

Treatment

The introduction of enzyme replacement therapy with recombinant human α -Gal A (agalsidase) has transformed the treatment of Fabry disease. RCTs showed that agalsidase administration for 5 to 6 months resulted in reduced plasma and urine Gb3; amelioration of neuropathic



Fig. 48.12 Angiokeratoma in Fabry Disease. Note the multiple periumbilical angiokeratomas. (Courtesy Dr. S. Waldek.)

pain; enhanced quality of life; clearing of Gb3 deposits from kidney, heart, and skin; and improved cerebral blood flow.⁴⁸ Treatment of classic Fabry disease patients with agalsidase- β with baseline proteinuria of less than 1 g/day had stabilization of GFR over 5 years of follow-up. Addition of RAS blockade to enzyme replacement can result in sustained reductions in proteinuria. In patients with ESKD, agalsidase may be infused during hemodialysis because there is little clearance of the enzyme by hemodialysis. Agalsidase therapy has been recommended for all affected males and symptomatic carrier females, but the drug is prohibitively expensive in many parts of the world.^{49–51} An alternative treatment approach based on chemical “chaperones” is under investigation; however, a recent study did not show a significant treatment effect.

Kidney transplantation is an effective treatment for advanced Fabry nephropathy but does not ameliorate the extrarenal manifestations. Transplanted kidneys from deceased donors or unaffected living donors may develop glycosphingolipid inclusions, but this usually is not clinically significant. Fabry heterozygotes should not become kidney donors. Coronary artery and cerebrovascular disease are the major causes of mortality in patients with Fabry disease who have undergone kidney transplantation. Kidney allograft recipients with Fabry disease are candidates for agalsidase treatment.

Fabry Disease in Childhood

Often it is not appreciated that the signs and symptoms of Fabry disease typically begin in childhood, particularly pain crises, acroparesthesias, angiokeratomas, and corneal opacities, and thus the diagnosis is frequently delayed until adulthood.⁵⁰ Accumulation of Gb3 in podocytes has been identified in children before onset of proteinuria; however, current recommendations do not support initiation of enzyme replacement therapy for asymptomatic children unless over age 16 years.⁴⁹ Symptomatic children with Fabry disease are potential candidates for agalsidase therapy.⁵⁰

NAIL-PATELLA SYNDROME

Definition

Nail-patella syndrome (NPS) is an uncommon autosomal dominant condition characterized by hypoplasia or absence of the patellae, dystrophic nails, dysplasia of the elbows and iliac horns, and kidney disease.

Etiology and Pathogenesis

NPS is caused by mutations in the LIM homeodomain transcription factor *LMX1B*, including missense, splicing, insertion or deletion, and nonsense alterations. In vitro studies of the transcriptional

effects of mixing wild-type and mutant *LMX1B* suggest that NPS results from haploinsufficiency of *LMX1B*, rather than a dominant-negative effect. *LMX1B* is essential for the maintenance of differentiated podocytes in adult kidneys through effects on actin cytoskeletal organization.

Clinical Manifestations

Kidney Involvement

Clinically apparent kidney disease occurs in fewer than half of NPS patients. The nephropathy is usually benign, with an approximately 3% to 5% risk for progression to ESKD.^{52,53} The clinical signs of NPS nephropathy appear in adolescence or young adulthood and typically include microhematuria and mild proteinuria, although some patients develop nephrotic syndrome and mild hypertension. The severity of these manifestations may differ substantially in related individuals.

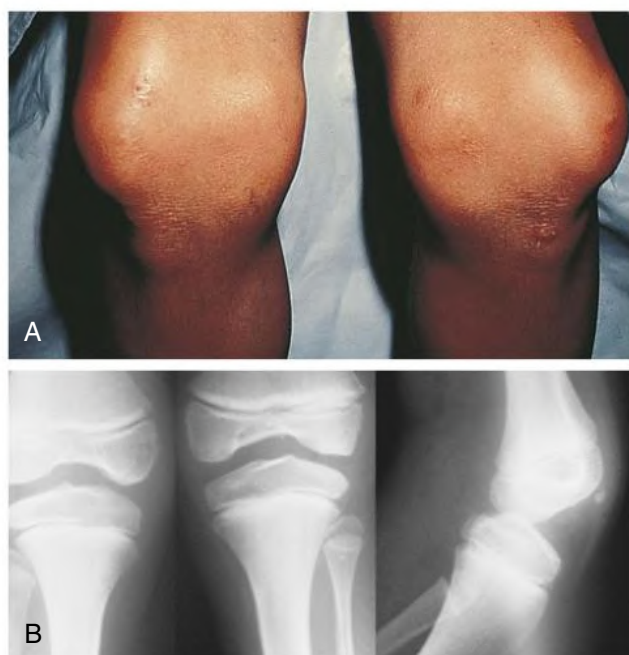


Fig. 48.13 Nail-patella Syndrome. Clinical (A) and radiologic (B) appearance of absence of the patellae. (Courtesy Dr. R. Vernier.)



Fig. 48.14 Nail-Patella Syndrome. Arrowheads denote iliac horns. ((Courtesy Dr. R. Vernier.))

Skeletal Defects

The patellae are absent or hypoplastic in more than 90% of patients with NPS, which may be associated with knee joint effusions and osteoarthritis (Fig. 48.13).^{52,53} In about 80% of patients, osseous processes project posteriorly from the iliac wings (iliac horns), which is pathognomonic (Fig. 48.14). Anomalies of the elbows include aplasia, hypoplasia, and posterior processes at the distal end of the humeri.

Nails

Nail abnormalities occur in about 98% of patients and are typically bilateral and symmetric. Fingernails are affected more often than toenails. The nails may be absent or dystrophic with discoloration, koilonychia, longitudinal ridges, or triangular lunulae.⁵²

Kidney pathology

LM findings in NPS are nonspecific and may demonstrate FSGS, mild mesangial hypercellularity, or no changes, in which case EM is required to make the diagnosis. EM shows multiple irregular lucencies of the GBM, giving it a moth-eaten appearance (Fig. 48.15). Such lucencies also may be observed in the mesangium. These lucent areas sometimes contain cross-banded collagen fibrils, which are more easily observed after staining with phosphotungstic acid (Fig. 48.16). The fibrils, which are type III collagen, tend to be arranged in clusters, and the surrounding GBM is often thickened. This may be observed without clinically evident kidney disease, but fibrils have not been found in extraglomerular basement membranes. Cross-banded fibrils of type III collagen have been seen in GBM of patients with glomerular disease who lack nail or skeletal abnormalities, sometimes as a familial condition with autosomal recessive inheritance (collagen III glomerulopathy; see Chapter 29). It is unclear whether there is a pathogenic relationship between collagen type III glomerulopathy and NPS.

Treatment

No specific therapy is available for the nephropathy of NPS. There has been no reported recurrence in transplanted kidneys. Because NPS is an autosomal dominant disorder, careful evaluation of potential living related kidney donors for features of NPS is essential.

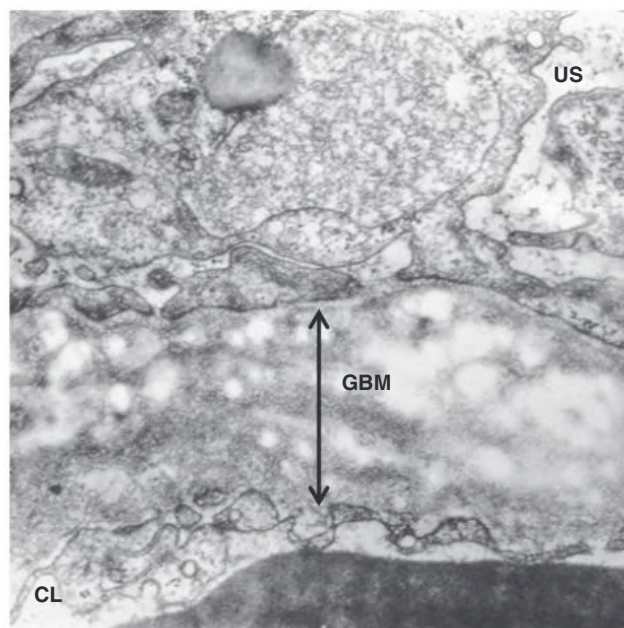


Fig. 48.15 Electron Micrograph of Kidney Biopsy Specimen in Nail-Patella Syndrome. The glomerular basement membrane (GBM) appears moth-eaten on routine staining. CL, Capillary lumen; US, urinary space. (Courtesy Dr. R. Vernier.)

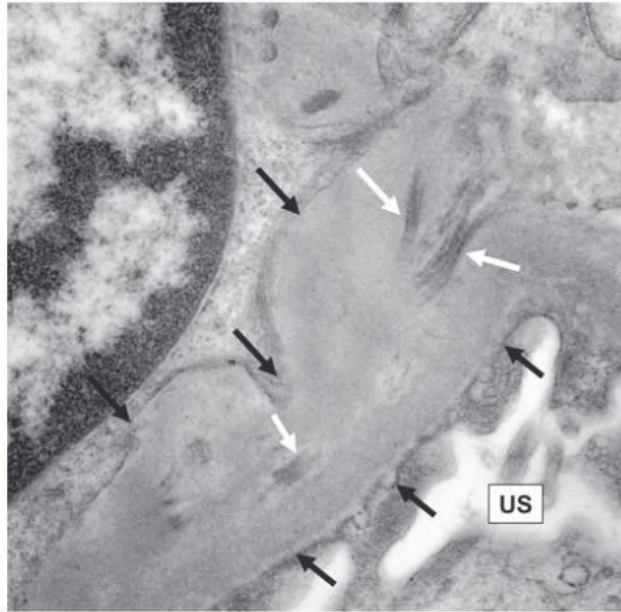


Fig. 48.16 Electron Micrograph of Kidney Biopsy Specimen in Nail-Patella Syndrome. *Black arrows* show margins of irregular glomerular basement membrane. Staining with phosphotungstic acid reveals fibrillar collagen (*white arrows*). *US*, Urinary space.

SELF-ASSESSMENT QUESTIONS

- A 15-year-old boy has AS. What statement about his parents is *most* likely to be *true*?

 - His father has AS.
 - His mother has AS.
 - Both parents are carriers of AS.
 - Neither parent has AS.
 - None of the above.
- Which of the following ocular abnormalities have been described in patients with AS?

 - Anterior lenticonus
 - Recurrent corneal erosions
 - Perimacular flecks
 - Posterior polymorphous dystrophy
 - All of the above
- Which one of the following statements about hearing loss in AS is true?

 - AS may be picked up by neonatal hearing screening.
 - In males with AS, hearing loss is usually undetectable by audiometry until adulthood.
 - Hearing loss in AS initially targets high frequencies above the range of conversational speech.
 - The hearing loss of AS is conductive.
 - Hearing loss in AS is typically unilateral.

REFERENCES

- Saran R, Li Y, Robinson B, et al. US Renal Data System 2015 Annual Data Report: Epidemiology of Kidney Disease in the United States. *Am J Kidney Dis*. 2016;67(Svii):S1–305.
- Khoshnoodi J, Pedchenko V, Hudson BG. Mammalian collagen IV. *Microsc Res Tech*. 2008;71:357–370.
- Fallerini C, Dosa L, Tita R, et al. Unbiased next generation sequencing analysis confirms the existence of autosomal dominant Alport syndrome in a relevant fraction of cases. *Clin Genet*. 2014;86:252–257.
- Jais JP, Knebelmann B, Giatras I, et al. X-linked Alport syndrome: natural history in 195 families and genotype-phenotype correlations in males. *J Am Soc Nephrol*. 2000;11:649–657.
- Yamamura T, Horinouchi T, Nagano C, et al. Genotype-phenotype correlation and the influence of the genotype on response to angiotensin-targeting drugs in Japanese patients with male X-linked Alport syndrome. *Kidney Int*. 2020.
- Jais JP, Knebelmann B, Giatras I, et al. X-linked Alport syndrome: natural history and genotype-phenotype correlations in girls and women belonging to 195 families: a “European Community Alport Syndrome Concerted Action” study. *J Am Soc Nephrol*. 2003;14:2603–2610.
- Rheault MN, Kren SM, Hartich LA, et al. X-inactivation modifies disease severity in female carriers of murine X-linked Alport syndrome. *Nephrol Dial Transplant*. 2010;25:764–769.
- Storey H, Savige J, Sivakumar V, Abbs S, Flinter FA. COL4A3/COL4A4 mutations and features in individuals with autosomal recessive alport syndrome. *J Am Soc Nephrol*. 2013.
- Kashtan CE, Ding J, Garosi G, et al. Alport syndrome: a unified classification of genetic disorders of collagen IV alpha345: a position paper of the Alport Syndrome Classification Working Group. *Kidney Int*. 2018;93:1045–1051.
- Kamiyoshi N, Nozu K, Fu XJ, et al. Genetic, clinical, and pathologic backgrounds of patients with autosomal dominant alport syndrome. *Clin J Am Soc Nephrol*. 2016;11:1441–1449.
- Marcocci E, Uliana V, Bruttini M, et al. Autosomal dominant Alport syndrome: molecular analysis of the COL4A4 gene and clinical outcome. *Nephrol Dial Transplant*. 2009;24:1464–1471.
- Voskarides K, Damianou L, Neocleous V, et al. COL4A3/COL4A4 mutations producing focal segmental glomerulosclerosis and renal failure in thin basement membrane nephropathy. *J Am Soc Nephrol*. 2007;18:3004–3016.
- Malone AF, Phelan PJ, Hall G, et al. Rare hereditary COL4A3/COL4A4 variants may be mistaken for familial focal segmental glomerulosclerosis. *Kidney Int*. 2014;86:1253–1259.
- Gast C, Pengelly RJ, Lyon M, et al. Collagen (COL4A) mutations are the most frequent mutations underlying adult focal segmental glomerulosclerosis. *Nephrol Dial Transplant*. 2016;31:961–970.
- Yao T, Udwan K, John R, et al. Integration of genetic testing and pathology for the diagnosis of adults with FSGS. *Clin J Am Soc Nephrol*. 2019;14:213–223.
- Merchant SN, Burgess BJ, Adams JC, et al. Temporal bone histopathology in alport syndrome. *Laryngoscope*. 2004;114:1609–1618.
- Streeten BW, Robinson MR, Wallace R, Jones DB. Lens capsule abnormalities in Alport's syndrome. *Arch Ophthalmol*. 1987;105:1693–1697.
- Temme J, Peters F, Lange K, et al. Incidence of renal failure and nephroprotection by RAAS inhibition in heterozygous carriers of X-chromosomal and autosomal recessive Alport mutations. *Kidney Int*. 2012;81:779–783.
- Grunfeld JP, Noel LH, Hafez S, Droz D. Renal prognosis in women with hereditary nephritis. *Clin Nephrol*. 1985;23:267–271.
- Savige J, Sheth S, Leys A, Nicholson A, Mack HG, Colville D. Ocular features in Alport syndrome: pathogenesis and clinical significance. *Clin J Am Soc Nephrol*. 2015;10:703–709.
- Savige J, Sheth S, Leys A, Nicholson A, Mack HG, Colville D. Ocular features in Alport syndrome: pathogenesis and significance. *Clin J Am Soc Nephrol*. 2015;10(4):703–709.
- Zhou J, Mochizuki T, Smeets H, et al. Deletion of the paired alpha 5(IV) and alpha 6(IV) collagen genes in inherited smooth muscle tumors. *Science*. 1993;261:1167–1169.
- Sa MJ, Fieremans N, de Brouwer AP, et al. Deletion of the 5'exons of COL4A6 is not needed for the development of diffuse leiomyomatosis in patients with Alport syndrome. *J Med Genet*. 2013;50:745–753.
- Heath KE, Campos-Barros A, Toren A, et al. Nonmuscle myosin heavy chain IIA mutations define a spectrum of autosomal dominant macrothrombocytopenias: May-Hegglin anomaly and Fechtner, Sebastian, Epstein, and Alport-like syndromes. *Am J Hum Genet*. 2001;69:1033–1045.
- Kashtan CE, Segal Y, Flinter F, Mäkanjuola D, Gan JS, Watnick T. Aortic abnormalities in males with Alport syndrome. *Nephrol Dial Transplant*. 2010;25:3554–3560.
- Kashtan CE, Gubler MC, Sisson-Ross S, Mauer M. Chronology of renal scarring in males with Alport syndrome. *Pediatr Nephrol*. 1998;12:269–274.
- Kovacs G, Kalmar T, Endreffy E, et al. Efficient targeted next generation sequencing-based workflow for differential diagnosis of alport-related disorders. *PLoS One*. 2016;11:e0149241.
- Savige J, Gregory M, Gross O, Kashtan C, Ding J, Flinter F. Expert guidelines for the management of Alport syndrome and thin basement membrane nephropathy. *J Am Soc Nephrol*. 2013;24:364–375.
- Kashtan CE, Kim Y. Distribution of the alpha 1 and alpha 2 chains of collagen IV and of collagens V and VI in Alport syndrome. *Kidney Int*. 1992;42:115–126.
- Kalluri R, Shield CF, Todd P, Hudson BG, Neilson EG. Isoform switching of type IV collagen is developmentally arrested in X-linked Alport syndrome leading to increased susceptibility of renal basement membranes to endoproteolysis. *J Clin Invest*. 1997;99:2470–2478.
- Kashtan CE, Kim Y, Lees GE, Thorner PS, Virtanen I, Miner JH. Abnormal glomerular basement membrane laminins in murine, canine, and human Alport syndrome: aberrant laminin alpha2 deposition is species independent. *J Am Soc Nephrol*. 2001;12:252–260.
- Dufek B, Meehan DT, Delimont D, et al. Endothelin A receptor activation on mesangial cells initiates Alport glomerular disease. *Kidney Int*. 2016;90:300–310.
- Kashtan CE, Gross O. Clinical practice recommendations for the diagnosis and management of Alport syndrome in children, adolescents, and young adults—an update for 2020. *Pediatr Nephrol*. 2020;36(3):711–719.
- Kashtan CE, Ding J, Gregory M, et al. Clinical practice recommendations for the treatment of Alport syndrome: a statement of the Alport Syndrome Research Collaborative. *Pediatr Nephrol*. 2013;28:5–11.
- Webb NJ, Lam C, Shahinfar S, et al. Efficacy and safety of losartan in children with Alport syndrome—results from a subgroup analysis of a prospective, randomized, placebo- or amlodipine-controlled trial. *Nephrol Dial Transplant*. 2011;26:2521–2526.
- Webb NJ, Shahinfar S, Wells TG, et al. Losartan and enalapril are comparable in reducing proteinuria in children with Alport syndrome. *Pediatr Nephrol*. 2013;28:737–743.
- Zhang Y, Wang F, Ding J, et al. Long-term treatment by ACE inhibitors and angiotensin receptor blockers in children with Alport syndrome. *Pediatr Nephrol*. 2016;31:67–72.
- Gross O, Licht C, Anders HJ, et al. Early angiotensin-converting enzyme inhibition in Alport syndrome delays renal failure and improves life expectancy. *Kidney Int*. 2012;81:494–501.
- Wuhl E, Trivelli A, Picca S, et al. Strict blood-pressure control and progression of renal failure in children. *N Engl J Med*. 2009;361:1639–1650.
- Gross O, Tonshoff B, Weber LT, et al. A multicenter, randomized, placebo-controlled, double-blind phase 3 trial with open-arm comparison indicates safety and efficacy of nephroprotective therapy with ramipril in children with Alport's syndrome. *Kidney Int*. 2020.
- Temme J, Kramer A, Jäger KJ, et al. Outcomes of male patients with Alport syndrome undergoing renal replacement therapy. *Clin J Am Soc Nephrol*. 2012;7:1969–1976.

41. Gross O, Weber M, Fries JW, Muller GA. Living donor kidney transplantation from relatives with mild urinary abnormalities in Alport syndrome: long-term risk, benefit and outcome. *Nephrol Dial Transplant*. 2009;24:1626–1630.
42. Plaisier E, Gribouval O, Alamowitch S, et al. COL4A1 mutations and hereditary angiopathy, nephropathy, aneurysms, and muscle cramps. *N Engl J Med*. 2007;357:2687–2695.
43. Matern D, Gavrilov D, Oglesbee D, Raymond K, Rinaldo P, Tortorelli S. Newborn screening for lysosomal storage disorders. *Semin Perinatol*. 2015;39:206–216.
44. Echevarria L, Benistan K, Toussaint A, et al. X-chromosome inactivation in female patients with Fabry disease. *Clin Genet*. 2016;89:44–54.
45. Orteu CH, Jansen T, Lidove O, et al. Fabry disease and the skin: data from FOS, the Fabry outcome survey. *Br J Dermatol*. 2007;157:331–337.
46. Sodi A, Ioannidis AS, Mehta A, Davey C, Beck M, Pitz S. Ocular manifestations of Fabry's disease: data from the Fabry Outcome Survey. *Br J Ophthalmol*. 2007;91:210–214.
47. Terryn W, Deschoenmakere G, De Keyser J, et al. Prevalence of Fabry disease in a predominantly hypertensive population with left ventricular hypertrophy. *Int J Cardiol*. 2013;167:2555–2560.
48. Schiffmann R, Kopp JB, Austin 3rd HA, et al. Enzyme replacement therapy in Fabry disease: a randomized controlled trial. *JAMA*. 2001;285:2743–2749.
49. Biegstraaten M, Arngrimsson R, Barbey F, et al. Recommendations for initiation and cessation of enzyme replacement therapy in patients with Fabry disease: the European Fabry Working Group consensus document. *Orphanet J Rare Dis*. 2015;10:36.
50. Hopkin RJ, Jefferies JL, Laney DA, et al. The management and treatment of children with Fabry disease: A United States-based perspective. *Mol Genet Metab*. 2016;117:104–113.
51. Schiffmann R, Hughes DA, Linthorst GE, et al. Screening, diagnosis, and management of patients with Fabry disease: conclusions from a “Kidney Disease: Improving Global Outcomes” (KDIGO) Controversies Conference. *Kidney Int*. 2017;91:284–293.
52. Ghoumid J, Petit F, Holder-Espinasse M, et al. Nail-Patella Syndrome: clinical and molecular data in 55 families raising the hypothesis of a genetic heterogeneity. *Eur J Hum Genet*. 2016;24:44–50.
53. Sweeney E, Fryer A, Mountford R, Green A, McIntosh I. Nail patella syndrome: a review of the phenotype aided by developmental biology. *J Med Genet*. 2003;40:153–162.